Improving Diffusion Inverse Problem Solving with Decoupled Noise Annealing

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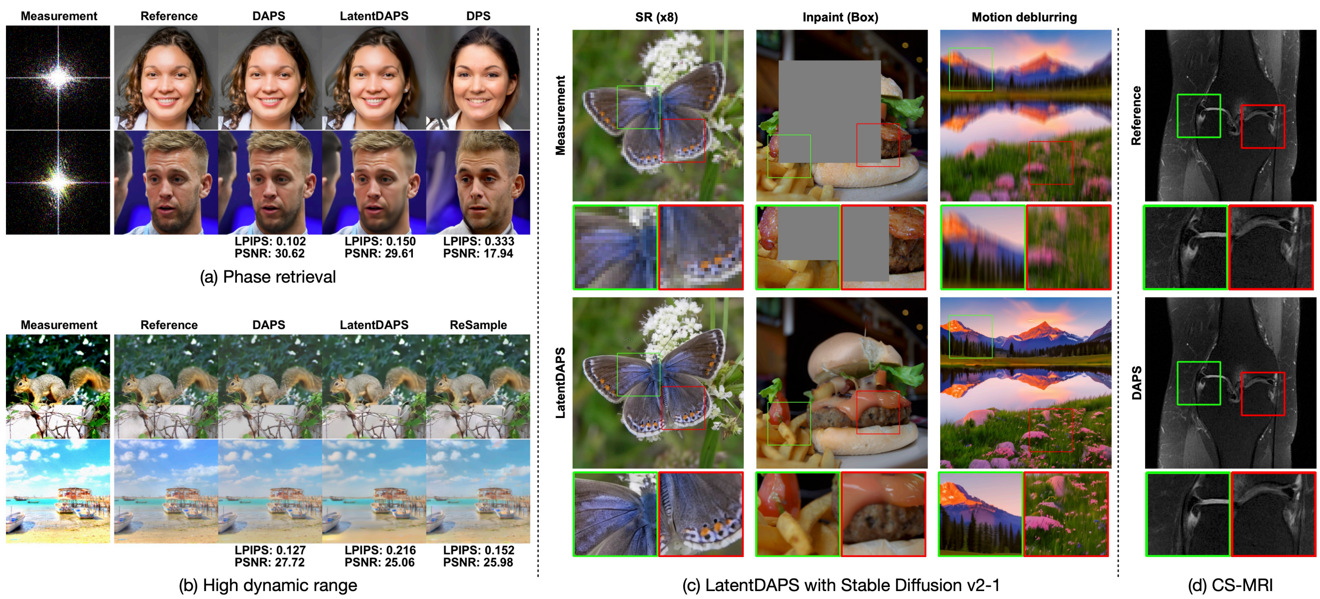

Figure 1. **Overview of Decoupled Annealing Posterior Sampling (DAPS).** Our method provides a flexible and effective framework for solving inverse problems through a decoupled posterior sampling process. In (a)(b), we present DAPS visual results on FFHQ and ImageNet at a resolution of 256, and in (c), on natural images at a resolution of 768. In (d), we display DAPS results on compressed sensing multi-coil MRI (CS-MRI). DAPS effectively addresses nonlinear inverse problems as well as medical imaging MRI challenges. Additionally, DAPS can be enhanced using large-scale latent diffusion models (LDMs) [40], as shown in (c).

Abstract

Diffusion models have recently achieved success in solving Bayesian inverse problems with learned data priors. Current methods build on top of the diffusion sampling process, where each denoising step makes small modifications to samples from the previous step. However, this process struggles to correct errors from earlier sampling steps, leading to worse performance in complicated nonlinear inverse problems, such as phase retrieval. To address this challenge, we propose a new method called Decoupled Annealing Posterior Sampling (DAPS) that relies on a novel noise annealing process. Specifically, we decouple consecutive steps in a diffusion sampling trajectory, allowing them to vary considerably from one another while ensuring their time-marginals anneal to the true posterior as we reduce noise levels. This approach enables the exploration of a larger solution space, improving the success rate for accurate reconstructions. We demonstrate that DAPS significantly improves sample quality and stability across multiple image restoration tasks, particularly in complicated nonlinear inverse problems.

1. Introduction

Inverse problems are ubiquitous in science and engineering, with applications ranging from image restoration [13, 30, 34, 44, 49, 59], medical imaging [10, 11, 17, 22, 23, 50] to astrophotography [1, 19, 51, 52]. Solving an inverse problem involves finding the underlying signal $\mathbf{x}_0$ from its partial, noisy measurement $\mathbf{y}$. Since the measurement process is typically noisy and many-to-one, inverse problems do not have a

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unique solution; instead, multiple solutions may exist that are consistent with the observed measurement. In the Bayesian inverse problem framework, the solution space is characterized by the posterior distribution $p(\mathbf{x}_0 \mid \mathbf{y}) \propto p(\mathbf{y} \mid \mathbf{x}_0)p(\mathbf{x}_0)$, where $p(\mathbf{y} \mid \mathbf{x}_0)$ represents the noisy measurement process, and $p(\mathbf{x}_0)$ is the prior distribution. In this work, we aim to solve Bayesian inverse problems where the measurement process $p(\mathbf{y} \mid \mathbf{x}_0)$ is known, and the prior distribution $p(\mathbf{x}_0)$ is captured by a deep generative model trained on a corresponding dataset.

As score-based diffusion models [21, 26, 27, 47–49] have risen to dominance in modeling high-dimensional data distributions like images, audio, and video, they have become the leading method for estimating the prior distribution $p(\mathbf{x}_0)$ in Bayesian inverse problems. A diffusion model generates a sample $\mathbf{x}_0$ by smoothly removing noise from an unstructured initial noise sample $\mathbf{x}_T$ through solving stochastic differential equations (SDEs). In particular, each step of the sampling process recursively converts a noisy sample $\mathbf{x}_{t+\Delta t}$, where $\Delta t > 0$ denotes the step size, to a slightly less noisy sample $\mathbf{x}_t$ until $t = 0$. This iterative structure in the diffusion sampling process can be leveraged to facilitate Bayesian inverse problem solving. In fact, prior research [5, 13, 46] has shown that given measurement $\mathbf{y}$ and a diffusion model prior $p(\mathbf{x}_0)$, we can sample from the posterior distribution $p(\mathbf{x}_0 \mid \mathbf{y})$ by perturbing a reverse-time SDE using an approximate gradient of the measurement process, $\nabla_{\mathbf{x}_t} \log p(\mathbf{y} \mid \mathbf{x}_t)$, at every step of the SDE solver.

Despite the remarkable success of these methods in solving many real-world inverse problems, like image colorization [30, 32, 49], image super-resolution [13, 30, 46, 49], computed tomography [45, 50], and magnetic resonance imaging [23, 50], they face significant challenges in more complex inverse problems with nonlinear measurement processes, such as phase retrieval and nonlinear motion deblurring. This is partially because, in prior methods, each denoising step approximately samples from the distribution $p(\mathbf{x}_t \mid \mathbf{x}_{t+\Delta t}, \mathbf{y})$. This causes $\mathbf{x}_t$ and $\mathbf{x}_{t+\Delta t}$ to be close to each other because of using a small step size Δt in discretizing the reverse-time SDE. As a result, $\mathbf{x}_t$ can at most correct local errors in $\mathbf{x}_{t+\Delta t}$ but struggles to correct global errors that require significant modifications to the prior sample. This issue is exacerbated when the methods are applied to complicated, nonlinear inverse problems, such as phase retrieval, where they often converge to undesired samples that are consistent with the measurement but reside in low-probability areas of the prior distribution.

To address this challenge, we propose a new framework for solving general inverse problems, termed **D**ecoupled **A**nnealing **P**osterior **S**ampling (DAPS). Our method employs a new noise annealing process inspired by the diffusion sampling process, where we decouple the consecutive samples $\mathbf{x}_{t+\Delta t}$ and $\mathbf{x}_t$ in the sampling trajectory, allowing samplers to correct large, global errors made in earlier steps. Instead of repetitively sampling from $p(\mathbf{x}_t \mid \mathbf{x}_{t+\Delta t}, \mathbf{y})$ as in previous methods, which restricts the distances between consecutive samples $\mathbf{x}_{t+\Delta t}$ and $\mathbf{x}_t$, DAPS recursively samples from the marginal distribution $p(\mathbf{x}_t \mid \mathbf{y})$. As illustrated in Fig. 2, we factorize the time marginal $p(\mathbf{x}_t \mid \mathbf{y})$ into three conditional distributions and sample from them in turn by solving the reverse diffusion process, simulating Langevin dynamics, and adding noise according to the forward diffusion process. We show that this creates approximate samples from corresponding marginal distributions. As the noise gradually anneals to zero, the time marginal $p(\mathbf{x}_t \mid \mathbf{y})$ smoothly converges to the posterior distribution $p(\mathbf{x}_0 \mid \mathbf{y})$, providing samples that approximately solve the Bayesian inverse problem.

Empirically, our method demonstrates significantly improved performance on various inverse problems compared to existing approaches, ranging from image restoration to compressed sensing MRI. Our method can be combined with diffusion models in both raw pixel space and learned latent space, which we refer to as DAPS and LatentDAPS, respectively. As shown in Fig. 1, both methods provide superior reconstructions with improved visual perceptual quality across a wide range of nonlinear inverse problems. Our approach exhibits remarkable stability and sampling quality, particularly for challenging nonlinear inverse problems. On the FFHQ 256 dataset, we achieve PSNR values of 30.72 dB and 27.12 dB for the phase retrieval and high dynamic range tasks, representing improvements of 1.98 dB and 4.39 dB over the previous state-of-the-art results, respectively. We apply DAPS to large-scale text-conditioned latent diffusion models, showing its scalability to high-resolution image restoration tasks. Beyond image restoration tasks, DAPS achieves a PSNR of 31.49 dB, showing a 2.70 dB improvement over previous diffusion-based methods. We also demonstrate how DAPS can be applied to inverse problems on categorical data with pre-trained discrete diffusion models. Additionally, DAPS performs well with a small number of neural network evaluations (approximately 100), striking a better balance between efficiency and sample quality.

2. Background

2.1. Diffusion Models

Diffusion models [21, 26, 47–49] generate data by reversing a predefined noising process. Let the data distribution be $p(\mathbf{x}_0)$. We can define a series of noisy data distributions $p(\mathbf{x}; \sigma)$ by adding Gaussian noise with a standard deviation of σ to the data. These form the time-marginals of a stochastic differential equation (SDE) [26], given by $\mathrm{d}\mathbf{x}_t = \sqrt{2\dot{\sigma}_t \sigma_t}\mathrm{d}\mathbf{w}_t$, where σ_t is the predefined noise schedule with $\sigma_0 = 0$ and $\sigma_T = \sigma_{\max}$, $\dot{\sigma}_t$ is the time derivative of σ_t, and $\mathbf{w}_t$ is a standard Wiener process. We use $\mathbf{x}_t$ interchangeably with $\mathbf{x}_{\sigma_t}$. For a sufficiently large $\sigma_{\max}$, the

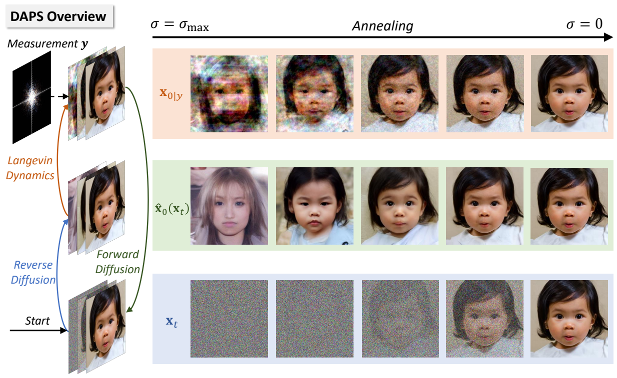
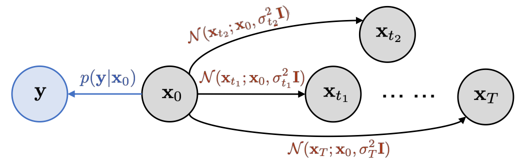
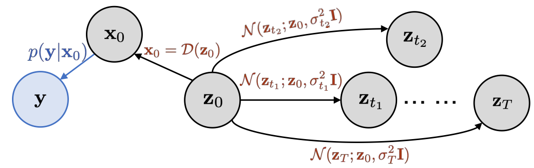


Figure 2. **An illustration of our method on phase retrieval**. Given a noisy sample $\mathbf{x}_t$, we first solve the reverse diffusion process to obtain $\hat{\mathbf{x}}_0(\mathbf{x}_t)$. Using this, we construct an approximation for $p(\mathbf{x}_0 \mid \mathbf{x}_t, \mathbf{y})$. Next, we sample $\mathbf{x}_{0|y} \sim p(\mathbf{x}_0 \mid \mathbf{x}_t, \mathbf{y})$ with multiple steps of Langevin dynamics, then perturb it with the forward diffusion process to obtain a sample with slightly less noise. We repeat this process until the noise reduces to zero.



(a) Diffusion models in pixel spaces.



(b) Latent diffusion models.

Figure 3. **Probabilistic graphical models** of the decoupled noise annealing process in (a) pixel and (b) latent spaces respectively. Here $\mathbf{x}_t \sim p(\mathbf{x}_t; \sigma_t)$, and $\mathbf{y}$ is the observed measurement. We factorize $p(\mathbf{x}_t \mid \mathbf{y})$ based on these probabilistic graphs.

distribution $p(\mathbf{x}; \sigma_{\max})$ converges to pure Gaussian noise $\mathcal{N}(\mathbf{0}, \sigma_{\max}^2 \boldsymbol{I})$.

To sample from data distribution $p(\mathbf{x}_0)$, we first draw an initial sample from $\mathcal{N}(\mathbf{0}, \sigma_{\max}^2 \boldsymbol{I})$, then solve the reverse SDE: $\mathrm{d}\mathbf{x}_t = -2\dot{\sigma}_t\sigma_t\nabla_{\mathbf{x}_t}\log p(\mathbf{x}_t; \sigma_t)\mathrm{d}t + \sqrt{2\dot{\sigma}_t\sigma_t}\mathrm{d}\mathbf{w}_t$, where $\nabla_{\mathbf{x}_t}\log p(\mathbf{x}_t; \sigma_t)$ is the time-dependent score function [47, 49]. Here $\nabla_{\mathbf{x}_t}\log p(\mathbf{x}_t; \sigma_t)$ is unknown, but can be approximated by training a diffusion model $\boldsymbol{s_\theta}(\mathbf{x}_t, \sigma_t)$ such that $\boldsymbol{s_\theta}(\mathbf{x}_t, \sigma_t) \approx \nabla_{\mathbf{x}_t}\log p(\mathbf{x}_t; \sigma_t)$.

2.2. Bayesian Inverse Problems with Diffusion

Inverse problems aim to recover data from partial, potentially noisy measurements. Formally, solving an inverse problem involves finding the inversion to a forward model that describes the measurement process. In general, a forward model takes the form of $\mathbf{y} = \mathcal{A}(\mathbf{x}_0) + \mathbf{n}$, where $\mathcal{A}$ is the measurement function, $\mathbf{x}_0$ represents the original data, $\mathbf{y}$ is the observed measurement, and $\mathbf{n}$ symbolizes the noise in the measurement process, often modeled as $\mathbf{n} \sim \mathcal{N}(\mathbf{0}, \beta_{\mathbf{y}}^2\boldsymbol{I})$. In a Bayesian framework, $\mathbf{x}_0$ comes from the posterior distribution $p(\mathbf{x}_0 \mid \mathbf{y}) \propto p(\mathbf{x}_0)p(\mathbf{y} \mid \mathbf{x}_0)$. Here $p(\mathbf{x}_0)$ is a prior distribution that can be estimated from a given dataset, and $p(\mathbf{y} \mid \mathbf{x}_0) = \mathcal{N}(\mathcal{A}(\mathbf{x}_0), \beta_{\mathbf{y}}^2\boldsymbol{I})$ models the noisy measurement process.

When the prior $p(\mathbf{x}_0)$ is modeled by a pre-trained diffusion model, we can modify reverse SDE to approximately sample from the posterior distribution following Bayes' rule,

$$\begin{aligned} \mathrm{d}\mathbf{x}_t = &- 2\dot{\sigma}_t\sigma_t\nabla_{\mathbf{x}_t}\log p(\mathbf{x}_t; \sigma_t)\mathrm{d}t \\ &- 2\dot{\sigma}_t\sigma_t\nabla_{\mathbf{x}_t}\log p(\mathbf{y} \mid \mathbf{x}_t)\mathrm{d}t + \sqrt{2\dot{\sigma}_t\sigma_t}\mathrm{d}\mathbf{w}_t. \end{aligned} \tag{1}$$

Here, the noisy likelihood $\nabla_{\mathbf{x}_t}\log p(\mathbf{y} \mid \mathbf{x}_t)$ is generally intractable. Multiple methods have been proposed to estimate the noisy likelihood [5, 8, 12, 36, 41, 46, 50, 54, 59]. One predominant approach is the DPS algorithm [13], which estimates $p(\mathbf{y} \mid \mathbf{x}_t) \approx p(\mathbf{y} \mid \mathbf{x}_0 = \mathbb{E}[\mathbf{x}_0 \mid \mathbf{x}_t])$. Another line of work [7, 18, 29, 30, 54] solves linear inverse problems by running the reverse diffusion in the spectral domain via singular value decomposition (SVD). Other methods bypass direct computation of this likelihood by interleaving optimization [2, 9, 24, 31, 45, 56, 59] or projection [10, 23, 30, 54] steps with normal diffusion sampling steps.

Despite promising empirical success, we find that this line of approaches faces challenges in solving more difficult inverse problems when the forward model is highly nonlinear. Accurately solving the reverse SDE in Eq. (1) requires the solver to take a very small step size $\Delta t > 0$, causing $\mathbf{x}_t$ and $\mathbf{x}_{t+\Delta t}$ to be very close to each other. Consequently, $\mathbf{x}_t$ can only correct minor errors in $\mathbf{x}_{t+\Delta t}$, but oftentimes fails to address larger, global errors that require substantial changes to $\mathbf{x}_{t+\Delta t}$. One such failure case is given in Fig. 4.

3. Method

3.1. Decoupled Annealing Posterior Sampling

Instead of solving the reverse-time SDE in Eq. (1), we propose a new noise annealing process that reduces the dependency between samples at consecutive time steps, as illustrated in Figs. 2, 3a and 3b. Unlike previous methods, we ensure $\mathbf{x}_t$ and $\mathbf{x}_{t+\Delta t}$ are conditionally independent given $\mathbf{x}_0$. To generate sample $\mathbf{x}_t$ from $\mathbf{x}_{t+\Delta t}$, we follow a two-step procedure: (1) sampling $\mathbf{x}_{0|\mathbf{y}} \sim p(\mathbf{x}_0 \mid \mathbf{x}_{t+\Delta t}, \mathbf{y})$, and (2) sampling $\mathbf{x}_t \sim \mathcal{N}(\mathbf{x}_{0|\mathbf{y}}, \sigma_t^2\boldsymbol{I})$. We repeat this process, gradually reducing noise until $\mathbf{x}_0$ is sampled. We call this process *decoupled noise annealing*, which is justified by the proposition below.

Proposition 1. *Suppose $\mathbf{x}_{t_1}$ is sampled from the time-marginal $p(\mathbf{x}_{t_1} \mid \mathbf{y})$, then*

$$\mathbf{x}_{t_2} \sim \mathbb{E}_{\mathbf{x}_0 \sim p(\mathbf{x}_0 \mid \mathbf{x}_{t_1}, \mathbf{y})}[\mathcal{N}(\mathbf{x}_0, \sigma_{t_2}^2\boldsymbol{I})] \tag{2}$$

satisfies the time-marginal $p(\mathbf{x}_{t_2} \mid \mathbf{y})$.

Remark. The key idea of Proposition 1 is to enable the sampling from $p(\mathbf{x}_{t_2} \mid \mathbf{y})$ for any noise level σ_{t_2} given any sample x_{t_1} at another noise level σ_{t_1}. For a sufficiently large σ_T, one can assume $p(\mathbf{x}_T \mid \mathbf{y}) \approx p(\mathbf{x}_T; \sigma_T) \approx \mathcal{N}(\mathbf{0}, \sigma_T^2 \boldsymbol{I})$. Starting from $\mathbf{x}_T$, we can iteratively sample from $p(\mathbf{x}_t \mid \mathbf{y})$ with σ_t annealed down from σ_T to 0.

The first step of our decoupled noise annealing requires sampling $\mathbf{x}_{0|\mathbf{y}} \sim p(\mathbf{x}_0 \mid \mathbf{x}_t, \mathbf{y})$ where $\mathbf{x}_t$ and $\mathbf{y}$ are known. Since $\mathbf{y}$ is conditionally independent from $\mathbf{x}_t$ given $\mathbf{x}_0$, we can deduce from the Bayes' rule that

$$
\begin{aligned}
p(\mathbf{x}_0 \mid \mathbf{x}_t, \mathbf{y}) &= \frac{p(\mathbf{x}_0 \mid \mathbf{x}_t) p(\mathbf{y} \mid \mathbf{x}_0, \mathbf{x}_t)}{p(\mathbf{y} \mid \mathbf{x}_t)} \\
&\propto p(\mathbf{x}_0 \mid \mathbf{x}_t) p(\mathbf{y} \mid \mathbf{x}_0). \qquad (3)
\end{aligned}
$$

To sample $\mathbf{x}_{0|\mathbf{y}}$ from this unnormalized distribution, MCMC methods such as Langevin dynamics [55] and Hamiltonian Monte Carlo [4] can be adopted (as explained in Appendix A.1). The updating rule of Langevin dynamics [55] is given by

$$
\begin{aligned}
\mathbf{x}_0^{(j+1)} &= \mathbf{x}_0^{(j)} + \eta \nabla_{\mathbf{x}_0^{(j)}} \log p(\mathbf{x}_0^{(j)} \mid \mathbf{x}_t) \\
&\quad + \eta \nabla_{\mathbf{x}_0^{(j)}} \log p(\mathbf{y} \mid \mathbf{x}_0^{(j)}) + \sqrt{2\eta} \boldsymbol{\epsilon}_j, \qquad (4)
\end{aligned}
$$

where $\eta > 0$ is the step size and $\boldsymbol{\epsilon}_j \sim \mathcal{N}(\mathbf{0}, \boldsymbol{I})$. When $\eta \to 0$ and $j \to \infty$, the sample $\mathbf{x}_0^{(j)}$ will be approximately distributed according to $p(\mathbf{x}_0 \mid \mathbf{x}_t, \mathbf{y})$.

3.2. Practical Design Choice of DAPS

Following Eq. (4), we discuss different ways to approximate the conditional distribution $p(\mathbf{x}_0 \mid \mathbf{x}_t)$ due to its intractability. A line of previous works on diffusion posterior sampling [5, 13, 46] approximate $p(\mathbf{x}_0 \mid \mathbf{x}_t)$ by a Gaussian distribution,

$$
p(\mathbf{x}_0 \mid \mathbf{x}_t) \approx \mathcal{N}(\mathbf{x}_0; \hat{\mathbf{x}}_0(\mathbf{x}_t), r_t^2 \boldsymbol{I}), \qquad (5)
$$

where $\hat{\mathbf{x}}_0(\mathbf{x}_t)$ is an estimator of $\mathbf{x}_0$ given $\mathbf{x}_t$ typically specified as $\mathbb{E}[\mathbf{x}_0 \mid \mathbf{x}_t]$, and the variance r_t^2 is specified using heuristics.

Another possible way to estimate $p(\mathbf{x}_0 \mid \mathbf{x}_t)$ is to decompose it via Bayes' rule, and substitute the intractable $p(\mathbf{x}_0)$ by the distribution $p(\mathbf{x}_0; \sigma_{t_{\min}})$ with a small Gaussian noise for numerical stability, i.e.,

$$
\begin{aligned}
\nabla_{\mathbf{x}_0} \log p(\mathbf{x}_0 \mid \mathbf{x}_t) &= \nabla_{\mathbf{x}_0} \log p(\mathbf{x}_t \mid \mathbf{x}_0) + \nabla_{\mathbf{x}_0} \log p(\mathbf{x}_0) \\
&\approx \nabla_{\mathbf{x}_0} \log p(\mathbf{x}_t \mid \mathbf{x}_0) + \boldsymbol{s}_{\boldsymbol{\theta}}(\mathbf{x}_0, t_{\min}). \qquad (6)
\end{aligned}
$$

Despite being a more accurate estimation, the *diffusion-score estimation* Eq. (6) is much more expensive than using a *Gaussian approximation* when applied to Eq. (4), as it requires calling the score function multiple times. Empirical studies in Sec. 4.4 show that using a *Gaussian approximation* can achieve comparable results as the *diffusion-score estimation*, but is significantly more time-efficient. In practice, we adopt the *Gaussian approximation* and compute $\hat{\mathbf{x}}_0(\mathbf{x}_t)$ by solving the (unconditional) probability flow ODE starting at $\mathbf{x}_t$. We leave details in Appendix F.2.

When the measurement noise is an isotropic Gaussian, *i.e.*, $\mathbf{n} \sim \mathcal{N}(\mathbf{0}, \beta_{\mathbf{y}}^2 \boldsymbol{I})$, the update rule simplifies to:

$$
\begin{aligned}
\mathbf{x}_0^{(j+1)} &= \mathbf{x}_0^{(j)} - \eta \nabla_{\mathbf{x}_0^{(j)}} \frac{\|\mathbf{x}_0^{(j)} - \hat{\mathbf{x}}_0(\mathbf{x}_t)\|^2}{2r_t^2} \\
&\quad - \eta \nabla_{\mathbf{x}_0^{(j)}} \frac{\|\mathcal{A}(\mathbf{x}_0^{(j)}) - \mathbf{y}\|^2}{2\beta_{\mathbf{y}}^2} + \sqrt{2\eta} \boldsymbol{\epsilon}_j, \qquad (7)
\end{aligned}
$$

Combining Eq. (7) with Proposition 1, we implement Decoupled Annealing Posterior Sampling (DAPS) as Algorithm 1 in Appendix A. In particular, given a noise schedule σ_t and a time discretization $\{t_i, i = 0, \ldots, N_A\}$, we iteratively sample $\mathbf{x}_{t_i}$ from the measurement-conditioned time-marginal $p(\mathbf{x}_{t_i} \mid \mathbf{y})$ for $i = N_A, N_A - 1, \ldots, 0$, following Eq. (2) in Proposition 1. This algorithm provides an approximate sample $\mathbf{x}_0$ from the posterior distribution $p(\mathbf{x}_0 \mid \mathbf{y})$.

The computational cost of Langevin dynamics mostly originates from evaluating the measurement function $\mathcal{A}$. In most image restoration tasks, this operation is significantly more efficient compared to evaluating the diffusion model. As demonstrated in Appendix E.1, the MCMC sampling introduces only a small overhead to the sampling process.

3.3. DAPS with Latent Diffusion Model

Given a pretrained encoder $\mathcal{E}$ and decoder $\mathcal{D}$, the latent diffusion models (LDMs) [40] are trained to fit the latent distribution $p(\mathbf{z}_0)$, where $\mathbf{z}_0$ is given by the encoder $\mathbf{z}_0 = \mathcal{E}(\mathbf{x}_0)$. One can recover $\mathbf{x}_0$ from the decoder $\mathbf{x}_0 = \mathcal{D}(\mathbf{z}_0)$. We found the proposed method can be adapted to sampling with pre-trained latent diffusion models by factorizing the probabilistic graphical model for the latent diffusion process, as shown in Fig. 3b. We refer to this as LatentDAPS in the following paper.

Accordingly, we can derive the following Langevin dynamics updating rule from Eq. (4):

$$
\begin{aligned}
\mathbf{z}_0^{(j+1)} &= \mathbf{z}_0^{(j)} + \eta \nabla_{\mathbf{z}_0^{(j)}} \log p(\mathbf{z}_0^{(j)} \mid \mathbf{z}_t) \\
&\quad + \eta \nabla_{\mathbf{z}_0^{(j)}} \log p(\mathbf{y} \mid \mathcal{D}(\mathbf{z}_0^{(j)})) + \sqrt{2\eta} \boldsymbol{\epsilon}_j. \qquad (8)
\end{aligned}
$$

While LatentDAPS is a straightforward and natural extension of DAPS compared to recent latent diffusion inverse problem solvers [42, 43, 45], it demonstrates comparable or even superior performance across several tasks in Tab. 1 and Tab. 4. More detailed derivation and proofs are shown in Appendix B.

3.4. Discussion with Existing Methods

Comparison with other posterior sampling methods. Unlike most previous algorithms, our sampling method does not

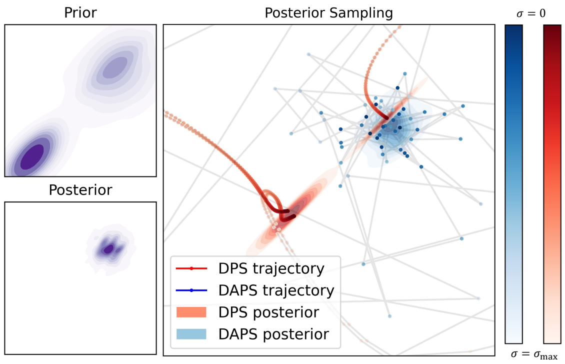


Figure 4. **DAPS vs. DPS on 2D synthetic data.** Consecutive sampling steps are close to each other for DPS but not for DAPS. Here DAPS approximates the posterior better than DPS.

solve a specific SDE/ODE; instead, we recursively sample from the time-marginal $p(\mathbf{x}_t \mid \mathbf{y})$ with noise annealing to zero. We therefore decouple the dependency of $\mathbf{x}_t$ and $\mathbf{x}_{t+\Delta t}$ in the sampling process. We argue that decoupling helps correct the errors accumulated in early diffusion sampling steps by allowing non-local transitions. This is particularly important when the measurement function is nonlinear. As a concrete example, Fig. 4 compares DAPS and DPS when solving a 2D nonlinear inverse problem with a Gaussian mixture prior. We visualize the trajectories of $\mathbf{x}_t$ from time T to 0 for both methods. For DAPS, points on the trajectory have significantly larger variations compared to those of DPS. As a result, DPS converges to wrong solutions, but DAPS is able to approximate the true posterior distribution. We provide more discussions in Appendix G.

Comparison with optimization-based methods. Although our method does not directly involve inner-loop optimization, we highlight that it connects with existing optimization-based solvers for inverse problems. For example, ReSample [45], DiffPIR [59], and DCDP [31] alternate between denoising optimizing, and resampling to solve inverse problems. However, these methods solve the MAP estimation problems, while DAPS draws samples from the posterior distribution. In particular, instantiating DAPS with Langevin dynamics and *Gaussian approximation* resembles [45] if we set standard deviation $\beta_{\mathbf{y}} \to 0$ and assume $\eta\beta_{\mathbf{y}}^2$ is constant in Eq. (18).

4. Experiments

4.1. Experimental Setup

We evaluate our method using both pixel-space and latent diffusion models. For pixel-based diffusion experiments, we leverage the pre-trained diffusion models trained by [13] on the FFHQ dataset and the pre-trained model from [16] on the ImageNet dataset. For latent diffusion models, we use the same pre-trained models as [45]: the unconditional LDM-VQ4 trained on FFHQ and ImageNet by [40]. The autoencoder's downsample factor is 4. We use the same time step discretization and noise schedule as EDM [26].

As mentioned in Sec. 3, we implement a few-step Euler ODE solver to compute $\hat{\mathbf{x}}_0(\mathbf{x}_t)$, while maintaining the same number of neural function evaluations (NFE) for different noise levels. In our experiments, we use DAPS-1K for all linear tasks and DAPS-4K for all nonlinear tasks. DAPS-1K uses 4 ODE solver NFE and 250 annealing scheduling steps, while DAPS-4K uses 10 NFE and 400 steps, respectively. We discuss the effect of choosing these configurations in Sec. 4.4. The corresponding latent version, LatentDAPS, follows the same settings but performs sampling in the latent space of VAEs. We use 100 and 50 Langevin steps per denoising iteration for DAPS and LatentDAPS respectively, and tune learning rates separately for each task. Further details on model configurations, samplers, and other hyperparameters are provided in Appendix F, along with more discussion on sampling efficiency in Sec. 4.2 and Appendix E.1.

Datasets and metrics. Adopting the previous convention, we test our method on two image datasets, FFHQ 256×256 [25] and ImageNet 256×256 [15]. To evaluate our method, we use 100 images from the validation set for both FFHQ and ImageNet. We include peak signal-to-noise-ratio (PSNR), structural similarity index measure (SSIM), Learned Perceptual Image Patch Similarity (LPIPS)[58] score and Fréchet inception distance (FID)[20] as our main evaluation metrics. For both our method and baselines, we use the versions implemented in piq [28] with all images normalized to the range $[0, 1]$. The replace-pooling option is enabled for LPIPS evaluation.

Inverse problems. We evaluate our method with a series of linear and nonlinear tasks. For linear inverse problems, we consider (1) super-resolution, (2) Gaussian deblurring, (3) motion deblurring, (4) inpainting (with a box mask), and (5) inpainting (with a 70% random mask). For Gaussian and motion deblurring, kernels of size 61×61 with standard deviations of 3.0 and 0.5, respectively, are used. In the super-resolution task, a bicubic resizer downscales images by a factor of 4. The box inpainting task uses a random box of size 128×128 to mask the original images, while random mask inpainting uses a generated random mask where each pixel has a 70% chance of being masked, following the settings in [45].

We consider three nonlinear inverse problems: (1) phase retrieval, (2) high dynamic range (HDR) reconstruction, and (3) nonlinear deblurring. Due to the inherent instability of phase retrieval, we adopt the strategy from DPS [13], using an oversampling rate of 2.0 and reporting the best result out of four independent samples. The goal of HDR reconstruction is to recover a higher dynamic range image (factor of 2) from a low dynamic range image. For nonlinear deblurring, we use the default setting as described in [53].

All linear and nonlinear measurements are subject to white Gaussian noise with a standard deviation of $\beta_{\mathbf{y}} = 0.05$. Further details regarding the forward measuring functions for each task and their respective hyperparameters are provided in the Appendix F.

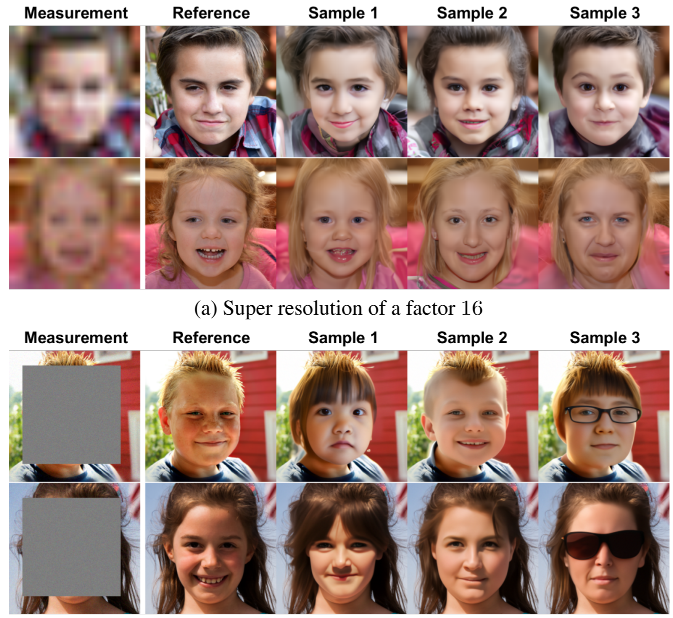
Baselines. We compare our methods with the following baselines: DDRM [30], DPS [13], DDNM [54], DCDP [31], FPS-SMC [18], DiffPIR [59] for pixel-based diffusion model experiments. We compare the latent diffusion version of our methods with PSLD [42] and ReSample [45], which also operate in the latent space. Note that DDRM, DDNM, FPS-SMC, and PSLD cannot handle nonlinear inverse problems. We especially RED-diff [35] for nonlinear experiments and a traditional method HIO [14] for phase retrieval.

4.2. Main Results

We show quantitative results for FFHQ and ImageNet dataset in Tab. 1. The qualitative comparisons are provided in Fig. 1. As shown in Tab. 1, our method achieves comparable or even better performance across all selected linear inverse problems. Moreover, our methods are remarkably stable when handling nonlinear inverse problems. For example, although existing methods such as DPS can recover high-quality images from phase retrieval measurements, they suffer from an extremely high failure rate, resulting in a relatively low PSNR and a high LPIPS. However, DAPS demonstrates superior stability across different samples and achieves a significantly higher success rate, resulting in a substantial improvement in PSNR and LPIPS as evidenced by the results in Tab. 1. Moreover, DAPS captures and recovers much more fine-grained details in measurements compared to existing baselines. We include more samples and comparisons in Appendix I for further illustration.

To understand the evolution of the noise annealing procedure, we evaluate two crucial trajectories in our method: 1) the estimated mean of $p(\mathbf{x}_0 \mid \mathbf{x}_t)$, denoted as $\hat{\mathbf{x}}_0(\mathbf{x}_t)$; and 2) samples from $p(\mathbf{x}_0 \mid \mathbf{x}_t, \mathbf{y})$ obtained through Langevin dynamics, denoted as $\mathbf{x}_{0|\mathbf{y}}$. We assess image quality and measurement error across these trajectories, as depicted in Fig. 7. The measurement error for $\mathbf{x}_{0|\mathbf{y}} \sim p(\mathbf{x}_0 \mid \mathbf{x}_t, \mathbf{y})$ remains consistently low throughout the process, while the measurement error for $\hat{\mathbf{x}}_0(\mathbf{x}_t)$ continuously decreases. Consequently, metrics such as PSNR and LPIPS exhibit stable improvement as the noise level is annealed.

In addition, our method can generate diversified samples when the measurements contain less information. To illustrate this, we selected two tasks where the posterior distributions may have multiple modes: (1) super-resolution by a factor of 16, and (2) box inpainting with a box size of 192×192. Some samples from DAPS are shown in Fig. 5, demonstrating DAPS's ability to produce diverse samples while preserving the measurement information. We show more samples in Appendix I for further demonstration.



(a) Super resolution of a factor 16

(b) Inpainting of 192×192 box

Figure 5. **Sample diversity**. We present several diverse samples generated by the DAPS under two sparse measurements whose posterior distributions contain multiple modes. DAPS produces a variety of samples with distinct features, including differences in expression, wearings, and hairstyles.

DAPS with large-scale text-conditioned latent diffusion. Due to the flexibility of DAPS, pretrained large-scale text-conditioned latent diffusion models (LDMs) [38–40] can also serve as the prior model $p(\mathbf{x}_0 \mid \mathbf{c})$ when given a text prompt $\mathbf{c}$. To demonstrate DAPS's potential, we use Stable Diffusion v2.1 [40] to present LatentDAPS results on three tasks with 768×768 resolution images, as shown in Fig. 1. Additionally, we provide the text prompts used, along with a detailed discussion, in Appendix B.1, where we also include a quantitative evaluation.

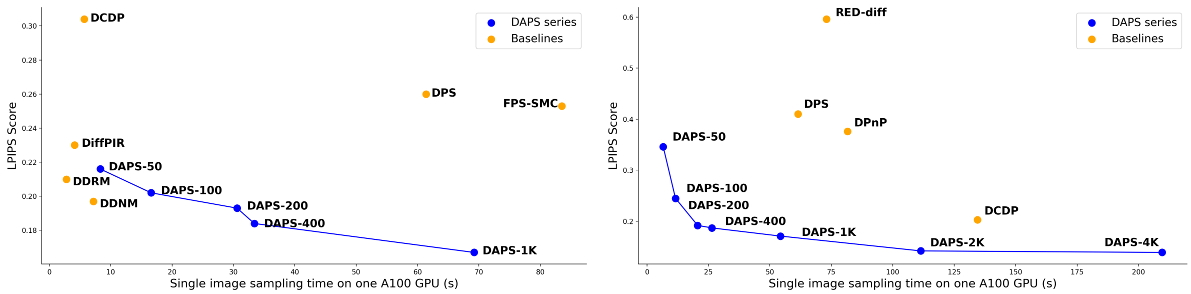
DAPS with discrete diffusion. DAPS can be naturally extended to support posterior sampling with categorical prior distribution modeled by discrete diffusion models [3, 6, 33]. Instead of sampling $\mathbf{x}_{0|\mathbf{y}}$ by Langevin MC or Hamiltonian MC in continuous space, we use the Metropolis-Hasting algorithm to replace Eq. (4). We validate DAPS with discrete diffusion on a quantized MNIST dataset. We leave the details to Appendix C.

4.3. Further results on MRI

We further validate the effectiveness of DAPS on compressed sensing multi-coil MRI, which is a medical imaging technique that helps reduce the scan time of MRI via subsampling. We train a diffusion model using the pipeline from [26] with fastMRI knee dataset [57]. Tab. 2 shows that DAPS outperforms existing diffusion-based baselines. We include more qualitative results in Appendix H.

Task	Type	Method	FFHQ				ImageNet			
			PSNR (↑)	**SSIM (↑)**	**LPIPS (↓)**	**FID (↓)**	**PSNR (↑)**	**SSIM (↑)**	**LPIPS (↓)**	**FID (↓)**
Super resolution 4×	*Pixel*	DAPS (ours)	**29.07**	**0.818**	**0.177**	51.44	**25.89**	0.694	**0.276**	**83.57**
		DPS	25.86	0.753	0.269	81.07	21.13	0.489	0.361	106.32
		DDRM	26.58	0.782	0.282	79.25	22.62	0.521	0.324	103.85
		DDNM	28.03	0.795	0.197	64.62	23.96	0.604	0.475	98.62
		DCDP	28.66	0.807	0.178	53.81	-	-	-	-
		FPS-SMC	28.42	0.813	0.204	**49.25**	24.82	**0.703**	0.313	97.51
		DiffPIR	26.64	-	0.260	65.77	23.18	-	0.371	106.32
	Latent	LatentDAPS(ours)	**27.48**	**0.801**	**0.182**	59.62	25.06	0.673	**0.276**	**84.37**
		PSLD	24.35	0.649	0.287	74.36	**25.42**	**0.694**	0.360	97.45
		ReSample	23.29	0.594	0.392	93.18	22.61	0.576	0.370	113.42
Inpaint (box)	*Pixel*	DAPS(ours)	24.07	0.814	**0.133**	**43.10**	21.43	0.725	0.214	109.85
		DPS	22.51	0.792	0.209	61.27	18.94	0.722	0.257	126.52
		DDRM	22.26	0.801	0.207	78.62	18.63	0.733	0.254	116.37
		DDNM	24.47	**0.837**	0.235	46.59	21.64	**0.748**	0.319	**103.97**
		DCDP	23.89	0.760	0.163	45.23	-	-	-	-
		FPS-SMC	**24.86**	0.823	0.146	48.34	**22.16**	0.726	**0.208**	111.58
	Latent	LatentDAPS(ours)	23.99	0.802	0.194	46.52	17.19	0.624	0.340	145.63
		PSLD	**24.22**	**0.813**	**0.158**	**43.02**	**20.10**	**0.694**	0.465	146.53
		ReSample	20.06	0.749	0.184	53.21	18.29	0.631	**0.262**	**127.84**
Inpaint (random)	*Pixel*	DAPS(ours)	**31.12**	**0.844**	**0.098**	**32.17**	28.44	0.775	**0.135**	**54.25**
		DPS	25.46	0.823	0.203	69.20	23.52	0.745	0.297	87.53
		DDNM	29.91	0.817	0.121	44.37	**31.16**	**0.841**	0.191	63.84
		DCDP	30.69	0.842	0.142	52.51	-	-	-	-
		FPS-SMC	28.21	0.823	0.261	61.23	24.52	0.701	0.316	79.12
	Latent	LatentDAPS(ours)	**30.71**	**0.813**	0.141	**36.41**	27.59	0.772	0.164	61.62
		PSLD	30.31	0.809	0.221	47.21	**31.30**	**0.783**	0.337	83.21
		ReSample	29.61	0.746	**0.140**	39.85	27.50	0.756	**0.143**	**59.87**
Gaussian deblurring	*Pixel*	DAPS(ours)	**29.19**	**0.817**	**0.165**	**53.33**	26.15	0.684	**0.253**	**75.68**
		DPS	25.87	0.764	0.219	79.75	20.31	0.598	0.397	116.42
		DDRM	24.93	0.732	0.239	92.43	21.26	0.564	0.443	146.89
		DDNM	28.20	0.804	0.216	57.83	**28.06**	**0.703**	0.278	81.43
		DCDP	27.50	0.699	0.304	86.43	-	-	-	-
		FPS-SMC	26.54	0.773	0.253	67.45	23.91	0.601	0.387	91.72
		DiffPIR	27.36	-	0.236	59.65	22.80	-	0.355	93.36
	Latent	LatentDAPS(ours)	**27.93**	**0.764**	**0.234**	**64.52**	25.05	0.668	0.345	78.51
		PSLD	23.27	0.631	0.316	89.51	25.86	0.688	0.390	91.39
		ReSample	26.39	0.714	0.255	71.69	**25.97**	**0.703**	**0.254**	**65.35**
Motion deblurring	*Pixel*	DAPS(ours)	**29.66**	**0.847**	**0.157**	**39.49**	**27.86**	**0.766**	**0.196**	**61.83**
		DPS	24.52	0.801	0.246	65.23	18.96	0.629	0.423	137.81
		DCDP	25.08	0.512	0.364	125.13	-	-	-	-
		FPS-SMC	27.39	0.826	0.227	48.32	24.52	0.647	0.326	87.43
		DiffPIR	26.57	-	0.255	65.78	24.01	-	0.366	94.63
	Latent	LatentDAPS(ours)	27.00	0.814	0.283	46.84	26.83	**0.745**	0.296	73.62
		PSLD	22.31	0.678	0.336	96.15	20.85	0.594	0.511	124.67
		ReSample	**27.41**	**0.823**	**0.198**	**44.72**	**26.94**	0.738	**0.227**	**66.89**
Phase retrieval	*Pixel*	DAPS(ours)	**30.63**$_{\pm 3.13}$	**0.851**$_{\pm 0.072}$	**0.139**$_{\pm 0.060}$	**42.71**	**25.78**$_{\pm 6.92}$	**0.743**$_{\pm 0.084}$	**0.254**$_{\pm 0.125}$	**82.67**
		DPS	17.64$_{\pm 2.97}$	0.441$_{\pm 0.129}$	0.410$_{\pm 0.090}$	104.52	16.81$_{\pm 3.61}$	0.427$_{\pm 0.143}$	0.447$_{\pm 0.099}$	197.54
		RED-diff	15.60$_{\pm 4.48}$	0.398$_{\pm 0.195}$	0.596$_{\pm 0.092}$	167.43	14.98$_{\pm 3.75}$	0.386$_{\pm 0.057}$	0.536$_{\pm 0.129}$	212.24
		DCDP	28.65$_{\pm 8.09}$	0.781$_{\pm 0.217}$	0.203$_{\pm 0.196}$	68.13	-	-	-	-
	Latent	LatentDAPS(ours)	**29.16**$_{\pm 3.55}$	**0.796**$_{\pm 0.089}$	**0.199**$_{\pm 0.078}$	**54.26**	**20.54**$_{\pm 6.41}$	**0.612**$_{\pm 0.114}$	**0.361**$_{\pm 0.150}$	**129.54**
		ReSample	21.60$_{\pm 8.10}$	0.648$_{\pm 0.154}$	0.406$_{\pm 0.224}$	84.32	19.24$_{\pm 4.21}$	0.618$_{\pm 0.146}$	0.403$_{\pm 0.174}$	130.47
	Classical	HIO	13.53$_{\pm 2.50}$	0.359$_{\pm 0.093}$	0.726$_{\pm 0.068}$	268.09	-	-	-	-
Nonlinear deblur	*Pixel*	DAPS(ours)	28.29$_{\pm 1.77}$	0.783$_{\pm 0.036}$	**0.155**$_{\pm 0.032}$	49.38	27.73$_{\pm 3.23}$	0.724$_{\pm 0.048}$	**0.169**$_{\pm 0.056}$	59.87
		DPS	23.39$_{\pm 2.01}$	0.623$_{\pm 0.082}$	0.278$_{\pm 0.060}$	91.31	22.49$_{\pm 3.20}$	0.591$_{\pm 0.101}$	0.306$_{\pm 0.081}$	101.41
		RED-diff	**30.86**$_{\pm 0.51}$	**0.795**$_{\pm 0.028}$	0.160$_{\pm 0.034}$	**43.84**	**30.07**$_{\pm 1.41}$	**0.754**$_{\pm 0.023}$	0.211$_{\pm 0.083}$	**51.22**
		DCDP	27.92$_{\pm 2.64}$	0.779$_{\pm 0.067}$	0.183$_{\pm 0.051}$	51.96	-	-	-	-
	Latent	LatentDAPS(ours)	28.11$_{\pm 1.75}$	0.713$_{\pm 0.041}$	0.235$_{\pm 0.049}$	53.63	25.34$_{\pm 3.44}$	0.615$_{\pm 0.057}$	0.314$_{\pm 0.080}$	76.73
		ReSample	**28.24**$_{\pm 1.69}$	0.742$_{\pm 0.039}$	**0.185**$_{\pm 0.039}$	**51.62**	**26.20**$_{\pm 3.71}$	**0.653**$_{\pm 0.064}$	**0.206**$_{\pm 0.057}$	**61.16**
High dynamic range	*Pixel*	DAPS(ours)	**27.12**$_{\pm 3.53}$	**0.752**$_{\pm 0.041}$	**0.162**$_{\pm 0.072}$	**42.97**	**26.30**$_{\pm 4.10}$	**0.717**$_{\pm 0.067}$	**0.175**$_{\pm 0.107}$	**64.19**
		DPS	22.73$_{\pm 6.07}$	0.591$_{\pm 0.141}$	0.264$_{\pm 0.156}$	112.82	19.23$_{\pm 2.52}$	0.582$_{\pm 0.082}$	0.503$_{\pm 0.106}$	146.23
		RED-diff	22.16$_{\pm 3.41}$	0.512$_{\pm 0.083}$	0.258$_{\pm 0.089}$	108.32	22.03$_{\pm 5.90}$	0.601$_{\pm 0.094}$	0.274$_{\pm 0.198}$	113.48
	Latent	LatentDAPS(ours)	**25.94**$_{\pm 2.87}$	**0.751**$_{\pm 0.056}$	0.223$_{\pm 0.080}$	74.83	23.64$_{\pm 4.10}$	0.609$_{\pm 0.053}$	0.269$_{\pm 0.099}$	93.51
		ReSample	25.65$_{\pm 3.57}$	0.732$_{\pm 0.059}$	**0.182**$_{\pm 0.085}$	**67.22**	**25.11**$_{\pm 4.21}$	**0.633**$_{\pm 0.049}$	**0.198**$_{\pm 0.089}$	**87.66**

Table 1. **Quantitative evaluation on FFHQ and ImageNet on 5 linear and 3 nonlinear tasks.** Performance comparison of different methods on FFHQ (left) and ImageNet (right) in the image domain. For linear tasks, we report the mean performance (PSNR, SSIM and LPIPS) across 100 validation images. We provide both the mean and standard deviation for nonlinear tasks, where results exhibit higher instability. The FID scores are evaluated with the same 100 validation images. The best and second-best results within each type of task are indicated by **bold** and underlined marks, respectively. DAPS demonstrates superior performance on most tasks, with LatentDAPS showing competitive results. All tasks are using noisy measurement with noise level $\beta_{\mathbf{y}} = 0.05$.



(a) Gaussian deblur (b) Phase retrieval

Figure 6. **Evaluation of time cost and sample quality**. The x-axis indicates the single image sampling time on an A100-SXM4-80GB GPU and y-axis shows the LPIPS. The evaluation uses 100 FFHQ images.

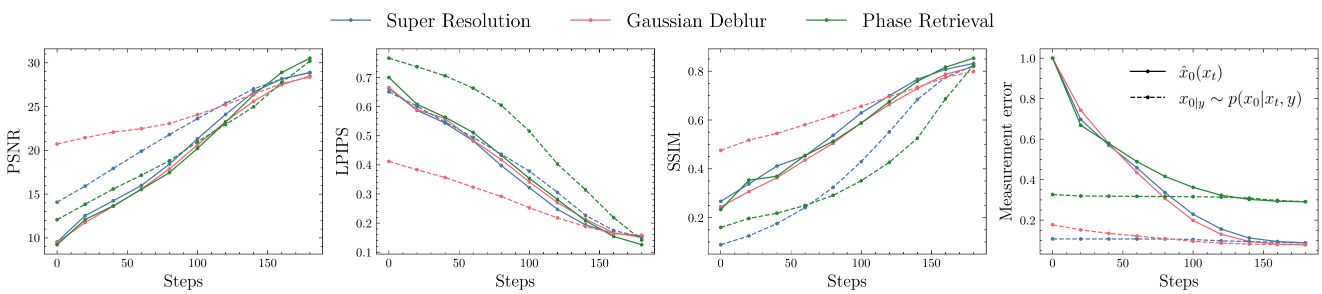


Figure 7. **Quantitative evaluations of image quality metrics and measurement error throughout the sampling process.** Here $\hat{\mathbf{x}}_0(\mathbf{x}_t)$ stands for the estimated mean of the Gaussian approximated distribution and $\mathbf{x}_{0|\mathbf{y}} \sim p(\mathbf{x}_0 \mid \mathbf{x}_t, \mathbf{y})$ means the samples from measurement conditioned time-marginal. The measurement errors are normalized between $[0, 1]$ for visualization.

	×4			×8		
	PSNR (↑)	SSIM (↑)	Meas err (↓)	PSNR (↑)	SSIM (↑)	Meas err (↓)
DAPS	**31.48**	**0.762**	1.57	**29.01**	**0.681**	1.28
DPS	26.13	0.620	9.90	20.82	0.536	6.74
DiffPIR	28.31	0.632	10.55	26.78	0.588	7.79
ScoreMRI [10]	25.97	0.468	10.83	25.01	0.405	8.36
CSGM [23]	28.78	0.710	**1.52**	26.15	0.625	**1.14**

Table 2. **Quantitative results of compressed sensing MRI.** We consider $\times 4$ and $\times 8$ subsampling ratio. Meas err stands for ℓ_2 data consistency error.

	Gaussian deblurring			High dynamic range		
	PSNR (↑)	LPIPS (↓)	Run time/s	PSNR (↑)	LPIPS (↓)	Run time/s
Gaussian approx.	29.40	0.170	64.7	24.85	0.181	61.4
diffusion-score	29.57	0.160	462.3	25.33	0.193	540.0
Relative improvement	−0.575%	−6.25%	614.53%	−1.89%	−6.22%	779.48%

Table 3. **Comparison of approaches to approximate $p(\mathbf{x}_0 \mid \mathbf{x}_t)$.** Three tasks on 10 FFHQ images with DAPS-1K consistently shown *Gaussian approximation* achieve huge speed-up while slightly sacrificing the performance compared with *diffusion-score estimation*.

4.4. Ablation Studies

Time efficiency of DAPS. We run DAPS with various configurations to test its performance under different computing budgets. Specifically, we evaluate DAPS with the number of function evaluations (NFE) of the diffusion models ranging from 50 to 4K, with configurations specified in Appendix E.1. Sec. 4.2 shows the relation between the sample quality of DAPS measured by LPIPS with its time cost, which indicates that DAPS is able to generate high-quality samples with a reasonable time cost.

Effectiveness of Different Design choices of DAPS. We conduct an ablation study to demonstrate the effectiveness of *Gaussian approximation* compared with the more principled but more expensive *diffusion-score estimation*. The results in Tab. 3 demonstrate that *Gaussian approximation* is $6 \sim 7$ times faster than *diffusion-score estimation* with the cost of a slight performance drop, effectively trading off performance for efficiency.

More ablation studies. We include more ablation studies of DAPS in Appendix I.1, including the number of function evaluations, the number of ODE steps, the annealing noise scheduling, and different measurement noise levels.

5. Conclusion

In summary, we propose Decoupled Annealing Posterior Sampling (DAPS) for solving inverse problems, particularly those with complex nonlinear measurement processes such as phase retrieval. Our method decouples consecutive sample points in a diffusion sampling trajectory, allowing them to vary considerably, thereby enabling DAPS to explore a larger solution space. Empirically, we demonstrate that DAPS generates samples with better visual quality and stability compared to existing methods when solving a wide range of challenging inverse problems.

Acknowledgments

We are grateful to Pika for providing the computing resources essential for this research. We also extend our thanks to the Kortschak Scholars Fellowship for supporting B.Z. and W.C. at Caltech. J.B. acknowledges support from the Wally Baer and Jeri Weiss Postdoctoral Fellowship. A.A. is supported in part by Bren endowed chair and by the AI2050 senior fellow program at Schmidt Sciences.

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